

SE3002 Assignment 01 Quality Evaluation Report

Assignment 01



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SE-C	Mashbot Campaign System
Application baseline	Mashbot baseline implemented with React, FastAPI, PostgreSQL, SQLAlchemy, Alembic, and Docker Compose.
Report Scope	Exactly 7 functional requirements and 3 non-functional requirements selected for this evaluation.

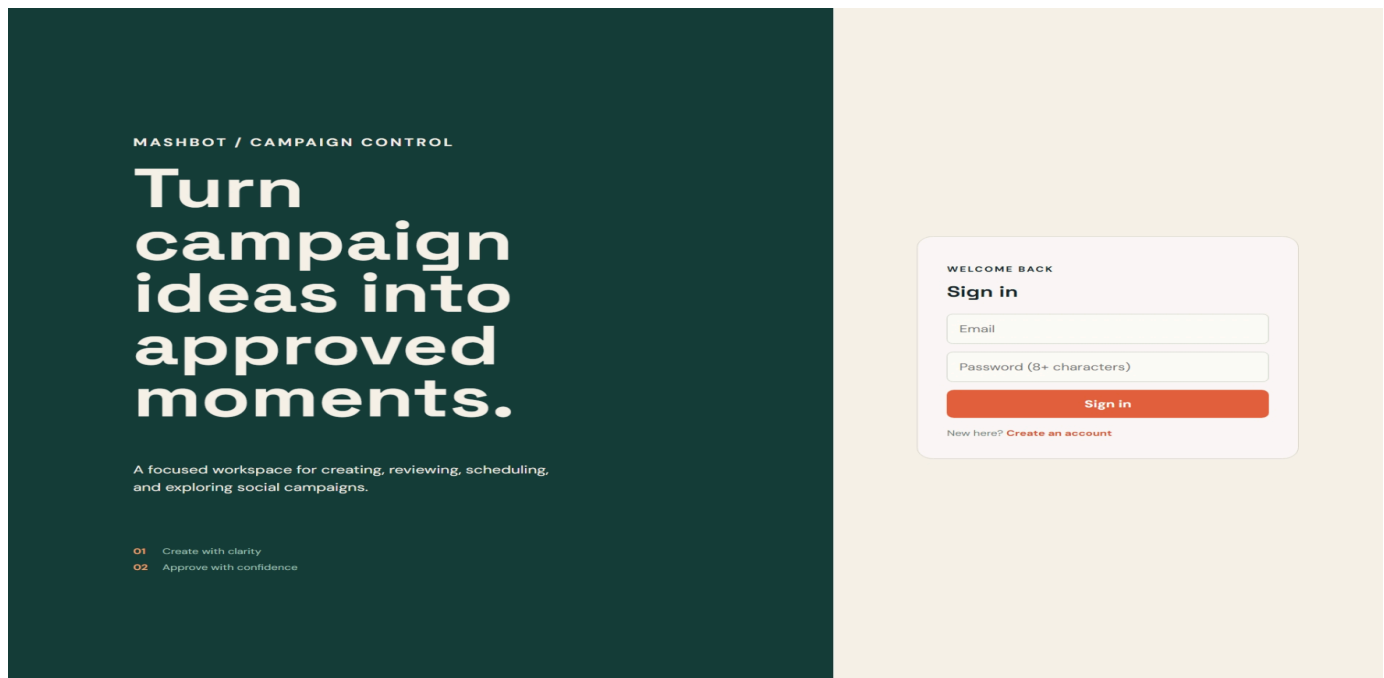


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1 Requirement Scope and AI Assumptions

The report evaluates only the selected Mashbot scope. Account management is treated as one coherent capability, role-based permissions are treated as a separate behaviour, and campaign approval and scheduling remain separate workflow requirements.

ID	Type	Requirement	Why selected and risk	AI assumption	Basis
0240, 0260-0370	FR1	Manage user accounts, including registration and authenticated profile modification	Combines account creation and modification into one complete user-account capability without counting related CRUD actions separately.	Every new account starts as a contributor; the administrator can later assign the required elevated role through People & roles.	SRS-supported
0650-0660	FR2	Authenticate valid users using username and password	Required before accessing campaigns, permissions, profile changes, and approval workflows.	Internal email/password authentication is sufficient for the selected scope.	Design decision
0140-0200	FR3	Apply role-based permissions to users and campaign actions	Adds meaningful authorization behaviour instead of another CRUD requirement.	The SRS roles are represented by contributor, approver, publisher, and admin permissions.	SRS-supported
0480-0550	FR4	Manage campaigns containing a name and text content	Provides the central business entity and a complete content-management workflow.	The selected scope supports text content only; image, audio, and video are deferred.	SRS-supported

0530	FR5	Schedule campaign content for publishing actions	Distinct from campaign management because scheduling controls when an approved action occurs.	Scheduling uses one configured application timezone because the SRS does not specify timezone behaviour.	Design decision
0210-0230, 0520	FR6	Process campaign content through contributor, approver, and publisher workflow	Adds meaningful state transitions and depends on RBAC without duplicating it.	Approval is required before publishing or scheduling; rejected content can be revised and resubmitted.	Design decision
0590-0610	FR7	Associate an account with an external service and interact through a standardized service interface	Preserves the external-service integration boundary without claiming unsupported live provider behaviour.	Provider associations are stored locally and the provider interface remains provider-neutral.	SRS-supported (boundary) Design decision(local storage only)
0620	NFR1	Encrypt communication between the web client and server	Priority 1 security requirement and independently evaluable through deployment inspection and HTTPS testing.	TLS will be provided through the Docker deployment boundary or reverse proxy.	Design decision
0630	NFR2	Support local or remote backup without unacceptable disruption	Priority 1 operational requirement and useful for SQE evidence.	The first implementation will demonstrate local PostgreSQL	SRS-supported (scope) Design Decision(local only)

				backup and restore; remote storage is deferred.	
0680	NFR3	Allow configurable automatic session timeout	Priority 1 security requirement and directly testable through session configuration and expiry policy.	Timeout is configured through environment/application settings rather than an administrator GUI.	SRS-supported (config location) Unsupported (60-min default value– no SRS threshold exists)

1.1 Requirement Dependency Analysis

Selected requirement	Depends on	Why the dependency matters
FR1: Account management	Database, validation, password hashing, authentication	Registration and profile modification require persistent identity data and an authenticated owner.
FR2: Authentication	Account registration, password hashing, session handling	Login cannot succeed without registered credentials and a session/token policy.
FR3: RBAC	Authentication, account role data	Permissions require a verified user and a stored role.
FR4: Campaign management	Authentication, RBAC, database	Campaign ownership and create permissions require an authenticated authorized user.
FR5: Scheduling	Campaign management, approval workflow, time validation	A schedule needs a campaign and must reject unapproved or past actions.
FR6: Approval workflow	Campaign management, RBAC	Submission and approval depend on campaign state and role permissions.
FR7: External service boundary	Authentication, account association, provider interface	Service associations must belong to a user and expose a stable interaction boundary.

NFR1: TLS	Docker deployment boundary or reverse proxy	Communication encryption is enforced outside the application routes.
NFR2: Backup	PostgreSQL persistence, backup command, restore target	Backup evidence requires a stable data store and a repeatable recovery process.
NFR3: Session timeout	Session/token handling and configuration	Timeout cannot be evaluated unless issued sessions carry an expiry policy.

2 AI Assisted Development Record

AI assistance was used to structure the React frontend, FastAPI backend, SQLAlchemy models, database migrations, Docker configuration, test cases, and this report. The pair remains responsible for verifying the implementation and evidence.

Assumption or decision	Basis	Verification status
Every newly registered account starts with the contributor role.	Explicit design decision aligned with least privilege and the selected workflow.	Verified by TC-01.
An administrator can later assign an elevated role such as publisher through People & roles.	SRS roles plus explicit role-management design decision.	Verified by TC-04; manual GUI evidence pending for TC-12.
Complete campaign CRUD is one FR, not separate create and modify FRs.	Explicit design decision aligned with assignment CRUD rule.	Implemented and tested.
Scheduling occurs only after approval; live third-party publishing is not claimed.	Explicit design decision to keep the workflow and provider boundary testable.	Implemented and tested at application boundary.
Only text campaign content is included in the selected scope.	SRS-supported scope decision; image, audio, and video are deferred.	Verified by campaign tests and documented limitation.

External-service associations are local provider-neutral records.	SRS-supported scope decision because real provider credentials/adapters are not configured.	Verified by TC-08; live provider behaviour is unsupported.
Local development admin account may use admin@mashbot.local.	Development configuration; output schema supports the local bootstrap account.	Verified through login and dashboard evidence.

2.1 Assumption Register

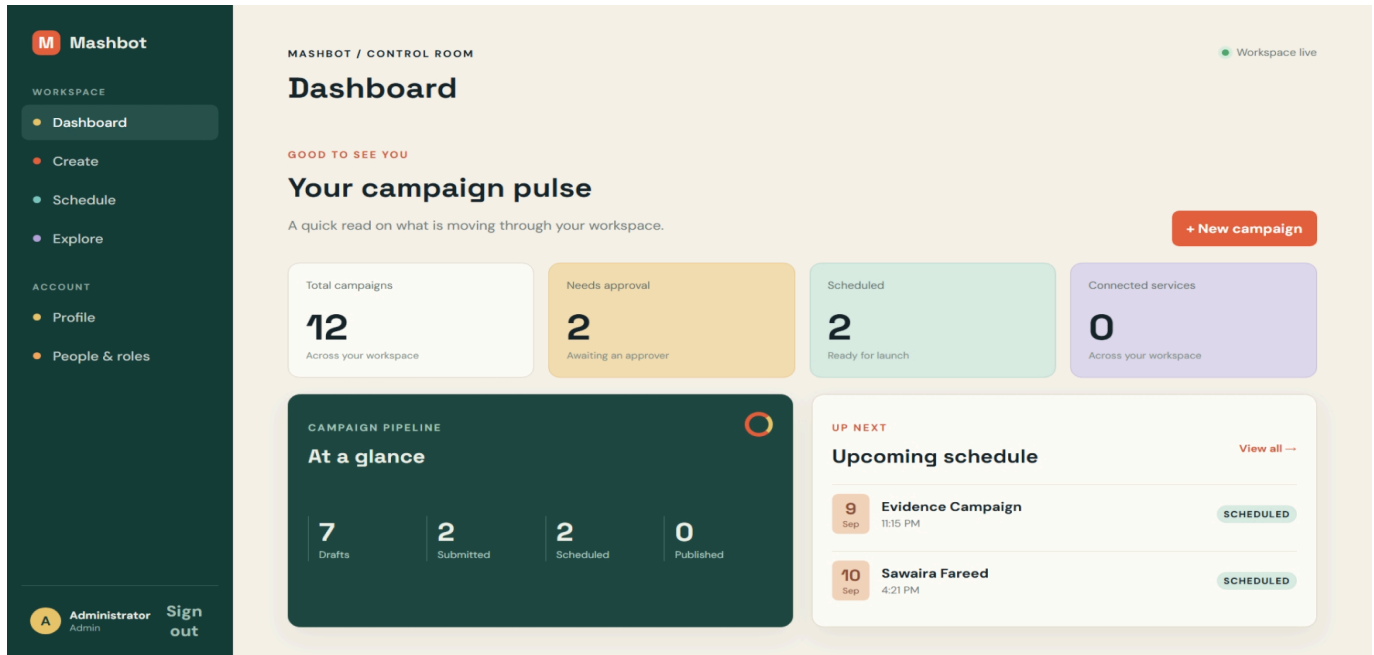
Assumption ID	Assumption	Basis	Impact if false	Status
A-01	New accounts are contributors by default.	Explicit design decision / least privilege.	Role-based workflow and TC-01 expectation change.	Implemented and verified.
A-02	Admin can later assign contributors to publishers through role management.	SRS role model and selected workflow.	Publisher scheduling workflow cannot be prepared.	Implemented; manual GUI evidence pending.
A-03	Approval is required before scheduling.	Selected workflow design decision.	Campaign lifecycle and TC-06/TC-07 expected results change.	Implemented and verified.
A-04	The external service is represented by a provider-neutral local boundary.	Scope decision due to unavailable real provider credentials.	Live publishing claims would be unsupported.	Implemented limitation.
A-05	A 60-minute token lifetime is an acceptable default.	Design decision; SRS requires configurability but gives no threshold.	Session-timeout judgment would need a different test basis.	Configuration verified; elapsed expiry not executed.

3 AI Generated GUI Baseline

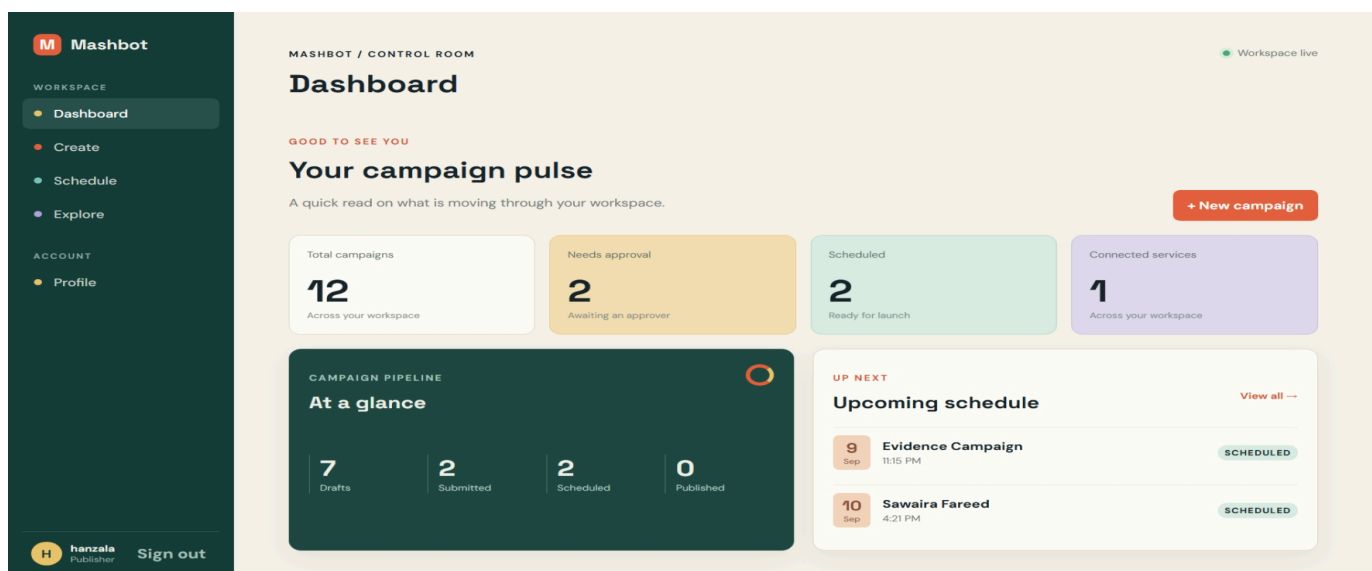
The GUI exposes Dashboard, Create, Schedule, Explore, Profile, and People & roles pages. The selected workflow is observable through login, role-controlled actions, campaign state changes, schedule entries, service-account associations, and keyword alerts.

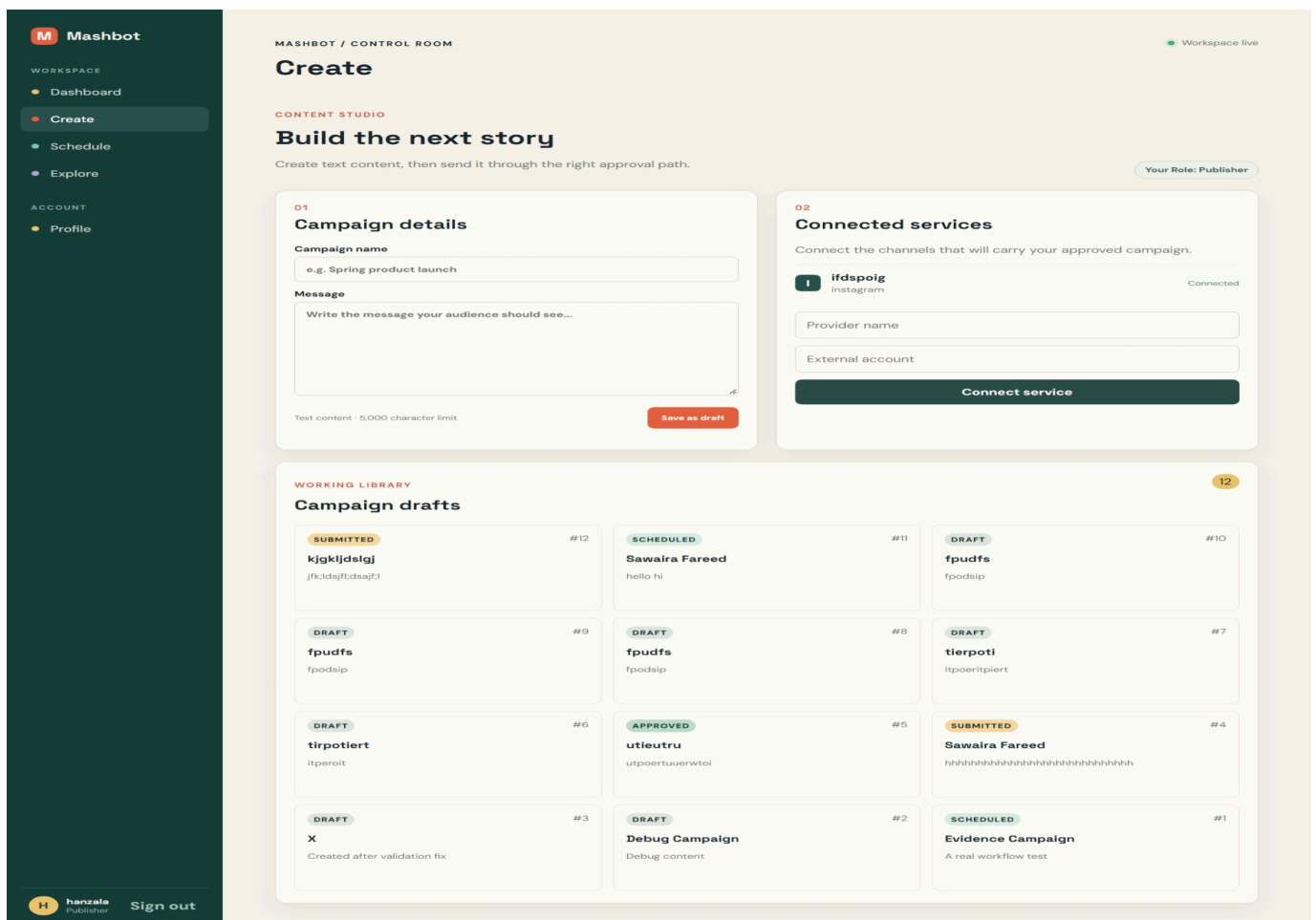
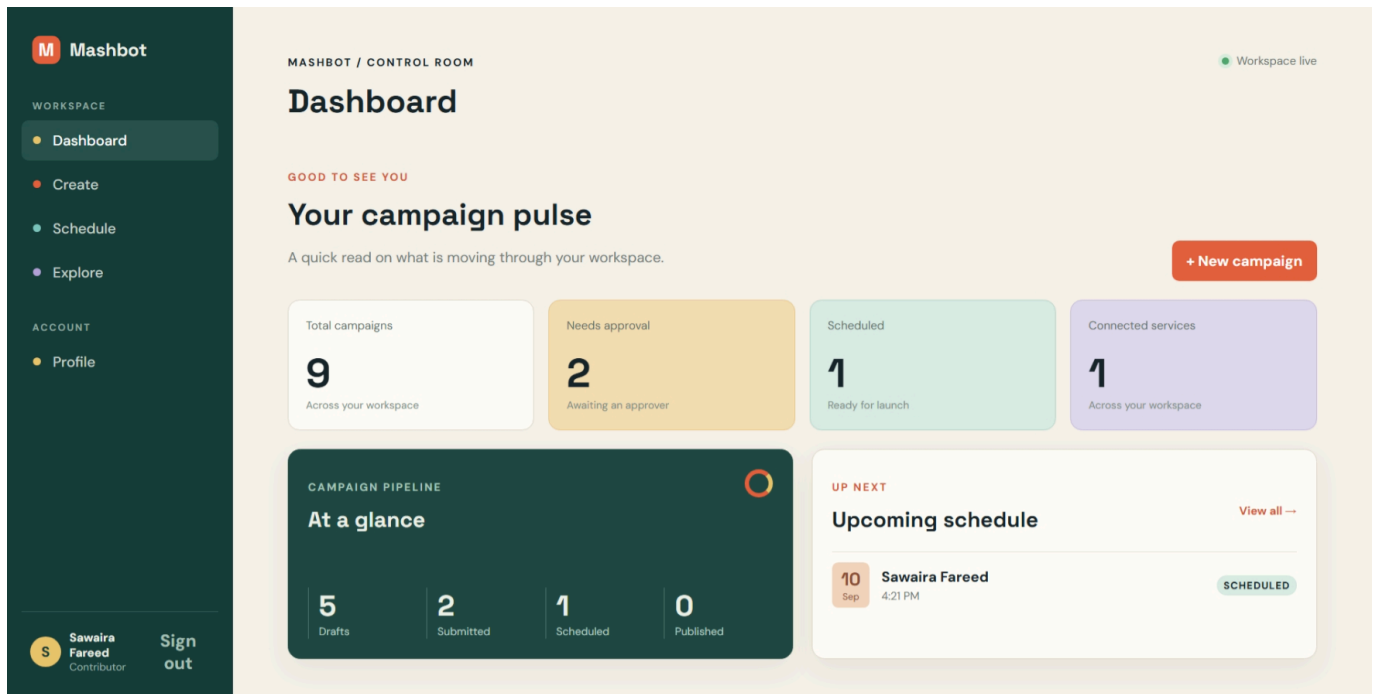
Run instructions: start Docker Desktop, run docker compose up --build from the project root, open <http://localhost:8080>, and use the API documentation at <http://localhost:8000/docs> when API evidence is needed.

Admin Dashboard :



Publisher Dashboard :





M Mashbot

WORKSPACE

Dashboard

Create

Schedule

Explore

ACCOUNT

Profile

H hanzala
Publisher

Sign out

MASHBOT / CONTROL ROOM

Workspace live

Schedule

LAUNCH CALENDAR

Schedule the moment

Only approved content can move onto the publishing calendar.

Choose launch time

mm/dd/yyyy --|-- --

READY TO GO

1

Approved campaigns

utieutru
utpoertuuerwtol

Schedule

CALENDAR FEED

Upcoming

10
Sep

Sawaira Fareed
9/10/2026, 4:21:00 PM

SCHEDULED

9
Sep

Evidence Campaign
9/9/2026, 11:15:01 PM

SCHEDULED

M Mashbot

WORKSPACE

Dashboard

Create

Schedule

Explore

ACCOUNT

Profile

People & roles

A Administrator
Admin

Sign out

MASHBOT / CONTROL ROOM

Workspace live

Explore

LISTENING POST

Explore the conversation

Keep a watchlist of topics that matter to your campaigns.

01

New keyword alert

Create a monitored search for a connected service.

Keyword or phrase

e.g. sustainable packaging

Service

All connected services

Create alert

YOUR WATCHLIST

1

Active signals

campaign

all · Monitoring

Pause

Provider data boundary

Keyword alerts are stored in Mashbot and ready for provider-specific search adapters. Live external-service results will appear here when a real provider integration is configured.

Page 11

Mashbot

WORKSPACE

Dashboard

Create

Schedule

Explore

ACCOUNT

Profile

People & roles

A Administrator Admin

Sign out

MASHBOT / CONTROL ROOM

Workspace live

Profile

YOUR IDENTITY

Profile settings

Keep your workspace identity and access details current.

Admin admin@mashbot.local

Display name

Administrator

New password

Leave blank to keep your password

Save profile

Mashbot

WORKSPACE

Dashboard

Create

Schedule

Explore

ACCOUNT

Profile

People & roles

A Administrator Admin

Sign out

MASHBOT / CONTROL ROOM

Workspace live

People & roles

ACCESS CONTROL

People & roles

Give each teammate the least privilege needed for their campaign work.

Admin Only

TEAM DIRECTORY

7 workspace members

A Administrator admin@mashbot.local

Admin

Active

T Test Contributor contributor1788891298@example.com

Contributor

Active

S Sawaira Fareed sawairafareed4@gmail.com

Contributor

Active

D Debug User debug1788949047@example.com

Contributor

Active

Q Quick User quick1788949215@example.com

Publisher

Active

H hanzala hanzala@gmail.com

Publisher

Active

S saba saba@gmail.cin

Contributor

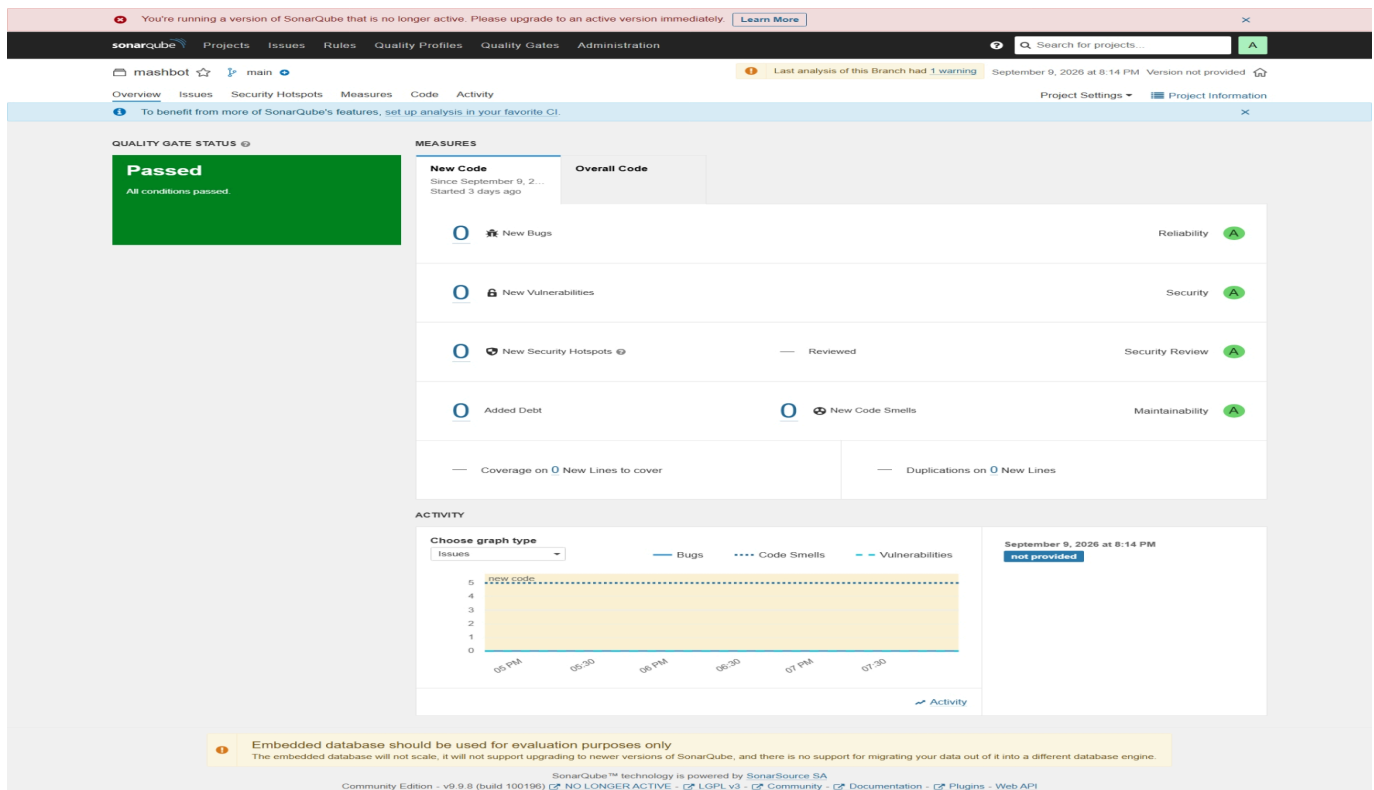
Active

Page 12

4 SonarQube Report and NFR Evaluation

The complete frozen codebase was scanned in the local SonarQube instance. The observed project metrics were: 5 code smells, 1 security hotspot, 0 bugs, 0 vulnerabilities, 0% coverage, 0% duplicated lines density, and 519 lines of code.

Quality area	Observed evidence	Interpretation
Maintainability	5 open code smells.	The baseline has maintainability findings in backend literals/default handling and frontend conditional rendering.
Reliability	0 bugs reported.	No SonarQube reliability bugs were reported in the scanned baseline.
Security	0 vulnerabilities and 1 security hotspot.	No vulnerability was reported; the hotspot requires review rather than being treated as a confirmed vulnerability.
Context	519 lines; 0% coverage; 0% duplication.	Coverage was not imported from a report, so 0% does not prove that no code is tested.





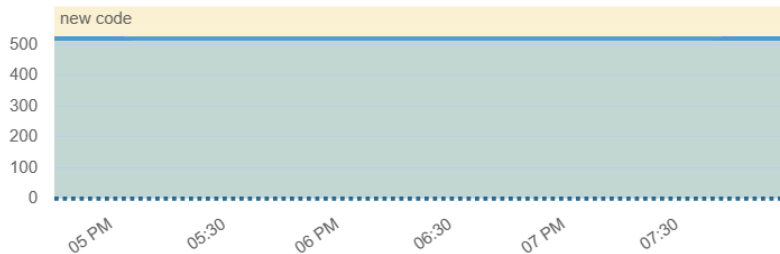
ACTIVITY

Choose graph type

Duplications

— Lines of Code

.... Duplicated Lines



September 9, 2026 at 8:14 PM

not provided

Activity

You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately.

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×

sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration

Search for projects...

A

Overview Issues Security Hotspots Measures Code Activity

Last analysis of this Branch had 1 warning

September 9, 2026 at 8:14 PM Version not provided

Project Settings Project Information

My Issues All

Bulk Change

1 / 5 issues 29min effort

Filters

Issues in new code

Type

Bug 0
Vulnerability 0
Code Smell 5

Severity

Blocker 0 Minor 0
Critical 3 Info 0
Major 2

Scope

Resolution

Status

Security Category

Creation Date

backend/app/main.py

Change this default value to "None" and initialize this parameter inside the function/method.

3 days ago L66

Code Smell Critical Open Not assigned 5min effort Comment

No tags

Define a constant instead of duplicating this literal "Campaign not found" 4 times.

3 days ago L116

Code Smell Critical Open Not assigned 8min effort Comment

design

backend/app/models.py

Define a constant instead of duplicating this literal "users.id" 3 times.

3 days ago L53

Code Smell Critical Open Not assigned 6min effort Comment

design

frontend/src/pages/DashboardPage.jsx

Extract this nested ternary operation into an independent statement.

3 days ago L9

Code Smell Major Open Not assigned 5min effort Comment

confusing

frontend/src/pages/SchedulePage.jsx

This statement will not be executed conditionally; only the first statement will be. The rest will execute unconditionally.

3 days ago L8

Code Smell Major Open Not assigned 5min effort Comment

cwe

5 of 5 shown



Embedded database should be used for evaluation purposes only

The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine.

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Where is the issue?

Why is this an issue?

mashbot backend/app/main.py See all issues in this file

```
65 @app.patch("/api/users/me", response_model=UserRead)
66 def update_profile(data: UserUpdate, db: Session = Depends(get_db), user: User = Depends(current_user)):
67
68     if data.name is not None:
69         1 user.name = data.name.strip()
70     if data.password is not None:
71         2 user.password_hash = hash_password(data.password)
72     db.commit()
73     db.refresh(user)
74     return user
```

Change this default value to "None" and initialize this parameter inside the function/method.

mashbot backend/app/main.py See all issues in this file

```
114 UserRole.admin)):
115     campaign = db.get(Campaign, campaign_id)
116     if not campaign:
117         raise HTTPException(status_code=404, detail="Campaign not found")
118
119     return submit_campaign(db, campaign, user)
120
121 @app.post("/api/campaigns/{campaign_id}/approve", response_model=CampaignRead)
122 def approve(campaign_id: int, db: Session = Depends(get_db), user: User = Depends(require_roles(UserRole.approver,
123     UserRole.admin)):
```

Define a constant instead of duplicating this literal "Campaign not found" 4 times.

mashbot backend/app/models.py See all issues in this file

```
50 content: Mapped[str] = mapped_column(Text)
51 status: Mapped[str] = mapped_column(String(30), default=CampaignStatus.draft.value)
52 scheduled_for: Mapped[datetime | None] = mapped_column(DateTime(timezone=True), nullable=True)
53 owner_id: Mapped[int] = mapped_column(ForeignKey("users.id"))
54
55 created_at: Mapped[datetime] = mapped_column(DateTime(timezone=True), default=utc_now)
56 owner: Mapped[User] = relationship(back_populates="campaigns")
57
58
59 class ServiceAccount(Base):
```

Define a constant instead of duplicating this literal "users.id" 3 times.

Code Smell Major Open Not assigned 5min effort 0 comments

confusing

Where is the issue?

Why is this an issue?

mashbot frontend/src/pages/DashboardPage.jsx See all issues in this file

```
{summary.upcoming.map((item) => <div className="upcoming-item" key={item.id}><div className="date-chip"><strong>{new
Date(item.scheduled_for).getDate()}</strong><small>{new Date(item.scheduled_for).toLocaleString([], { month: "short" })}
</small></div><div><strong>{item.name}</strong><p>{new Date(item.scheduled_for).toLocaleString([], { hour: "numeric", minute:
"2-digit" })}</p></div><span className="status scheduled">scheduled</span></div></div></article></section></div>;
```

Extract this nested ternary operation into an independent statement.

Code Smell ▾ Major ▾ Open ▾ Not assigned ▾ 5min effort 0 comments

cwe ▾

Where is the issue?

Why is this an issue?

frontend/src/pages/SchedulePage.jsx

```
~/campaigns/${id}/schedule`, { method: "POST", headers: authHeaders(token), body: JSON.stringify({ scheduled for: new Date(
date).toISOString() }) }); onMessage("Campaign scheduled."); setDate(""); refresh(); } catch (err) { onError(err.message); }
}
```

This statement will not be executed conditionally; only the first statement will be. The rest will execute unconditionally.

```
9 const ready = campaigns.filter((campaign) => campaign.status === "approved");
10 return <div className="page-stack"><section className="split-heading"><div><p className="eyebrow orange">LAUNCH
CALENDAR</p><h2>Schedule the moment</h2><p className="muted">Only approved content can move onto the publishing calendar.</p>
</div><label className="date-picker">Choose launch time<input type="datetime-local" value={date} onChange={(event) =>
setDate(event.target.value)} /></label></section><div className="schedule-grid"><section className="panel1"><div
```

Filters

Assigned to me All

Status

To review ▾

Overall code ▾

Security Hotspots Reviewed 0.0%

1 Security Hotspots to review

Review priority: LOW

Encryption of Sensitive Data 1

Using http protocol is insecure. Use https instead

backend/app/config.py

1 of 1 shown

Status: TO REVIEW

This security hotspot needs to be reviewed to assess whether the code poses a risk.

Change status ▾

Assignee: Not assigned

Where is the risk?

What's the risk?

Assess the risk

How can I fix it?

backend/app/config.py

Open in IDE

Get Permalink

```
1 from pydantic_settings import BaseSettings, SettingsConfigDict
2
3
4 class Settings(BaseSettings):
5     database_url: str = "postgresql+psycopg://mashbot:mashbot@db:5432/mashbot"
6     secret_key: str = "change-me-in-development"
7     access_token_minutes: int = 60
8     cors_origins: str = "http://localhost:5173,http://localhost:8080"
```

Using http protocol is insecure. Use https instead

Comment

```
9 bootstrap_admin_email: str | None = None
10 bootstrap_admin_password: str | None = None
```

Comment

You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately [Learn More](#)

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Search for projects...

Overview Issues Security Hotspots Measures Code Activity

Last analysis of this Branch had 1 warning September 9, 2026 at 8:14 PM Version not provided

Project Overview

Reliability

Security

Security Review

Maintainability

Coverage

Duplications

Size

Complexity

Issues

New Issues

Issues

Open Issues

Reopened Issues

Confirmed Issues

False Positive Issues

Won't Fix Issues

mashbot

View as List

to select files to navigate 19 files

Issues 5

New Code: Since September 9, 2026

backend/app/main.py	2
frontend/src/pages/DashboardPage.jsx	1
backend/app/models.py	1
frontend/src/pages/SchedulePage.jsx	1
backend/app/___init___py	0
frontend/src/api.js	0
frontend/src/components/AppShell.jsx	0
frontend/src/pages/AuthPage.jsx	0
backend/app/config.py	0
frontend/src/pages/CreatePage.jsx	0
backend/app/db.py	0
frontend/src/pages/ExplorePage.jsx	0
frontend/src/main.jsx	0
frontend/src/pages/PeoplePage.jsx	0
frontend/src/pages/ProfilePage.jsx	0
backend/app/schemas.py	0
backend/app/security.py	0
backend/app/services.py	0
backend/tests/test_health.py	0

19 of 19 shown

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▼ Reliability ?

Overview	
On new code	
Bugs	0
Rating	A
Remediation Effort	0
Overall	
Bugs	0
Rating	A
Remediation Effort	0

▼ Security ? SECURITY RATING

Overview	
On new code	
Vulnerabilities	0
Rating	A
Remediation Effort	0
Overall	
Vulnerabilities	0
Rating	A
Remediation Effort	0

Overview	
On new code	
Code Smells	0
Debt	0
Debt Ratio	0.0%
Rating	A
Overall	
Code Smells	5
Debt	29min
Debt Ratio	0.2%
Rating	A
Effort to Reach A	0

▼ Security Review ?

On new code	
Security Hotspots	0
Rating	A
Overall	
Security Hotspots	1
Rating	E
Security Hotspots Reviewed	0.0%

TypeCODE SMIELClear

Bug0

Vulnerability0

Code Smell5

Press Ctrl to add to selection

Severity

Blocker0

Critical3

Major2

Minor0

Info0

Severity

Blocker0

Critical3

Major2

Minor0

Info0

ScopeMAIN CODEClear

Main code5

Test code0

Press Ctrl to add to selection

Resolution

Unresolved5

Fixed0

Won't Fix0

False Positive0

Removed0

Status

Open5

Reopened0

Closed0

Confirmed0

Resolved0

Language

Search for languages...

JavaScript1

1 shown

Rule(JAVASCRIPT) TERNA...Clear

Search for rules...

(Python) String literals should not be dupl...2

(JavaScript) Multiline blocks should be e...1

(JavaScript) Ternary operators should no...1

(Python) Function parameters' default val...1

	Lines of Code	Bugs	Vulnerabilities	Code Smells	Security Hotspots	Coverage	Duplications
mashbot							
backend	375	0	0	3	1	0.0%	0.0%
frontend/src	144	0	0	2	0	0.0%	0.0%

2 of 2 shown

Page 18

4.1 NFR Evidence Plan

NFR ID	Evidence method	Evidence to attach	Current result	Limitation
0620	HTTPS request/redirect check and Docker deployment inspection	Screenshot Added	FAILED: HTTP 200 without TLS enforcement.	No TLS reverse proxy or certificate is configured.
0630	Controlled backup and restore inspection	ScreenShot added	FAILED: no backup/restore artefact is configured.	Restore cannot be executed until a backup mechanism exists.
0680	Configuration inspection plus token expiry test	Screenshot added	PASSED at configuration level; 60-minute default.	Full elapsed-time expiry was not run.

NFR	SonarQube evidence if relevant	Other evaluation method	Finding / judgment	Limitation
NFR1 TLS	N/A	Pytest transport check and Docker port inspection	FAILED: HTTP 200 was returned without HTTPS enforcement.	TLS termination is not configured in the local Docker deployment.
NFR2 Backup	N/A	Deployment inspection and backup artifact check	FAILED: no backup job, destination, retention policy, or restore procedure exists.	A restore test cannot be completed until a backup mechanism is supplied.
NFR3 Session timeout	N/A	Configuration and token-policy inspection	PASSED at configuration level: access token lifetime is configurable with a 60-minute default.	A full elapsed-time expiry test was not run.

5 Functional Test Derivation

Test conditions were derived for all seven selected functional requirements. The test basis is the selected SRS behaviour together with the explicit design decisions recorded above.

Test basis / requirement	Condition ID	Test condition
0240, 0260-0370	COND-01	A new account is created as contributor and an authenticated user can modify permitted profile data.
0650-0660	COND-02	Valid credentials authenticate; malformed, non-existent, missing, and invalid credentials are rejected.
0140-0200	COND-03	An administrator can assign a role and protected campaign actions enforce the assigned permissions.
0480-0550	COND-04	A permitted user can create text content at the exact 5,000-character boundary.
0530	COND-05	Only approved campaigns can be scheduled and past times are rejected.
0210-0230, 0520	COND-06	Submission and approval follow the permitted campaign state transitions.
0590-0610	COND-07	Valid external-service associations are stored and invalid boundary values are rejected.

6 Test Case Records

The following records use the assignment Table B fields. Automated API cases are evidenced by pytest output. System-level cases are intentionally left for manual browser execution and screenshot insertion.

TC-01 Valid account registration

Level / category	Functional/API, normal
Test basis / objective	FR1 (0240, 0260-0370); verify valid account creation and default role.
Preconditions	Email is unused.
Test data	Valid email, 2-character name, 8-character password.
Steps	Submit registration.
Expected result	HTTP 201; contributor account returned without password.
Actual result	HTTP 201 and default contributor role returned.
Status	PASSED
Evidence	pytest assertion.

TC-02 Authenticated profile modification

Level / category	Functional/API, normal
Test basis / objective	FR1 (0330-0370); verify an authenticated user can modify permitted profile data.
Preconditions	Registered user is authenticated.
Test data	New display name: Updated Profile.
Steps	PATCH /api/users/me with the authenticated token.
Expected result	HTTP 200 and updated name is returned.
Actual result	HTTP 200 and updated name returned.
Status	PASSED
Evidence	pytest assertion from profile modification check.

TC-03 Valid authentication

Level / category	Functional/API, normal
Test basis / objective	FR2 (0650-0660); verify registered user sign-in.
Preconditions	Active account exists.
Test data	Registered email and correct password.
Steps	Submit OAuth2 login form.
Expected result	HTTP 200 and bearer token returned.
Actual result	HTTP 200 and non-empty token returned.
Status	PASSED
Evidence	pytest assertion.

TC-04 Role assignment and enforcement

Level / category	Functional/security, normal and error
Test basis / objective	FR3 (0140-0200); verify admin role assignment and publisher restriction.
Preconditions	Admin and target contributor exist.
Test data	publisher role; publisher attempts campaign creation.
Steps	Assign role, then submit contributor-only create request.
Expected result	Role changes; forbidden action returns HTTP 403.
Actual result	Role changed and create request returned HTTP 403.
Status	PASSED
Evidence	pytest assertions.

TC-05 Maximum campaign content

Level / category	Functional/boundary
Test basis / objective	FR4 (0480-0550); verify exact 5,000-character text-content limit.
Preconditions	Permitted authenticated user exists.
Test data	Content of exactly 5,000 characters.
Steps	Create campaign.
Expected result	HTTP 201 and draft stored.
Actual result	HTTP 201 and draft stored.
Status	PASSED
Evidence	pytest assertion.

TC-06 Approval state transition

Level / category	Functional/workflow
Test basis / objective	FR6 (0210-0230, 0520); verify draft submission and approval.
Preconditions	Draft and approver exist.
Test data	Draft campaign.
Steps	Submit as owner, then approve as approver.
Expected result	Status changes to be submitted, then approved.
Actual result	Expected transitions occurred.
Status	PASSED
Evidence	pytest assertions.

TC-07 Invalid scheduling boundary

Level / category	Functional/error/boundary
Test basis / objective	FR5 (0530); verify past schedule time rejection.

Preconditions	Approved campaign exists.
Test data	Timestamp one second in the past.
Steps	Request scheduling.
Expected result	HTTP 422; campaign remains approved.
Actual result	HTTP 422; campaign was not scheduled.
Status	PASSED
Evidence	pytest assertion.

TC-08 External-service boundary validation

Level / category	Functional/boundary/error
Test basis / objective	FR7 (0590-0610); verify minimum valid values and invalid data.
Preconditions	Authenticated user exists.
Test data	provider=FB, external_username=u1; then provider=x.
Steps	Submit valid boundary payload and invalid provider payload.
Expected result	Valid payload returns 201; invalid payload returns 422.
Actual result	201 and 422 respectively.
Status	PASSED
Evidence	pytest assertions.

TC-09 Invalid and non-existent email handling

Level / category	Functional/invalid/error
Test basis / objective	FR2 (0650-0660); verify malformed registration and unregistered login rejection.
Preconditions	Test addresses are unused.

Test data	Malformed email; syntactically valid but non-existent email.
Steps	Submit malformed registration, then login with unregistered email.
Expected result	HTTP 422 and HTTP 401 respectively.
Actual result	Both requests were rejected as expected.
Status	PASSED
Evidence	pytest assertions.

TC-10 New user to dashboard journey

Level / category	System/manual
Test basis / objective	FR1, FR2, FR4; verify integrated GUI journey.
Preconditions	Docker services running; browser available.
Test data	New user credentials and campaign text.
Steps	Register, sign in, create draft, open Dashboard.
Expected result	The user sees the dashboard and creates a draft.
Actual result	User can create the campaign successfully.
Status	Passed
Evidence	Screenshot registration, dashboard, and created draft

01

Campaign details

Campaign name

Winter Sale

Message

I am going to launch my new winter Sale ! 🎉🎉 Yayyy

Text content - 5,000 character limit

Save as draft

02

Connected services

Connect the channels that will carry your approved campaign.

Winter Collections 2026

Instagram's

Instagram

Connect service

Campaign created as a draft.

DRAFT

#45

Winter Sale

I am going to launch my new winter Sale ! 🎉🎉 Yayyy

Submit for approval

Campaign submit completed.

02

Connected services

Connect the channels that will carry your approved campaign.

Winter Collections 2026

Instagram's

Instagram

Connect service



Winter Collections 2026
Instagram

Connected



Winter Collection Mega
TikTok

Connected

TC-11 End-to-end approval and scheduling

Level / category	System/manual
Test basis / objective	FR4, FR5, FR6; verify integrated multi-role workflow.
Preconditions	Contributor, approver, and publisher accounts exist.
Test data	Draft campaign and future launch time.
Steps	Create, submit, approve, sign in as publisher, schedule.
Expected result	Campaign appears as scheduled in the calendar.
Actual result	Campaigns scheduled successfully
Status	Passed
Evidence	Screenshot each role and calendar

Contributor :

WORKING LIBRARY

9

Campaign drafts

SUBMITTED

#12

kjgkljdsigj

jfk;ldsajf;dsajf;l

SCHEDULED

#11

Sawaira Fareed

hello hi

DRAFT

#10

fpudfs

fpodsip

Submit for approval

DRAFT

#9

fpudfs

fpodsip

Submit for approval

DRAFT

#8

fpudfs

fpodsip

Submit for approval

DRAFT

#7

tierpoti

itpoeritpiert

Submit for approval

DRAFT

#6

tirpotiert

itperolt

Submit for approval

APPROVED

#5

utietru

utpoertuerwtol

SUBMITTED

#4

Sawaira Fareed

hhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhh

Publisher :

h

hanzala

Publisher

Sign out

WORKING LIBRARY

13

Campaign drafts

SUBMITTED

#45

Winter Sale

I am going to launch my new winter Sale ! 🎉 🎉 Yayyy

APPROVE

REJECT

SUBMITTED

#12

kjgkljds!gkj

jfk;lds!f;dsajf;l

APPROVE

REJECT

SCHEDULED

#11

Sawaira Fareed

hello hi

APPROVE

REJECT

DRAFT

#10

fpudfs

fpodsip

APPROVE

REJECT

DRAFT

#9

fpudfs

fpodsip

APPROVE

REJECT

DRAFT

#8

fpudfs

fpodsip

APPROVE

REJECT

DRAFT

#7

tierpoti

itpoeritpiert

APPROVE

REJECT

DRAFT

#6

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itperoit

APPROVE

REJECT

APPROVED

#5

utlieutru

utpoertuuerwtol

APPROVE

REJECT

SUBMITTED

#4

Sawaira Fareed

hhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhh

APPROVE

REJECT

DRAFT

#3

X

Created after validation fix

APPROVE

REJECT

DRAFT

#2

Debug Campaign

Debug content

APPROVE

REJECT

SCHEDULED

#1

Evidence Campaign

A real workflow test

APPROVE

REJECT

Admin :

A

Administrator

Admin

Sign out

WORKING LIBRARY

13

Campaign drafts

SUBMITTED

#45

Winter Sale

I am going to launch my new winter Sale ! 🎉 🎉 Yayyy

APPROVE

REJECT

SUBMITTED

#12

kjgkljds!gkj

jfk;lds!f;dsajf;l

APPROVE

REJECT

SCHEDULED

#11

Sawaira Fareed

hello hi

APPROVE

REJECT

DRAFT

#10

fpudfs

fpodsip

APPROVE

REJECT

DRAFT

#9

fpudfs

fpodsip

APPROVE

REJECT

DRAFT

#8

fpudfs

fpodsip

APPROVE

REJECT

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APPROVE

REJECT

APPROVED

#5

utlieutru

utpoertuuerwtol

APPROVE

REJECT

SUBMITTED

#4

Sawaira Fareed

hhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhh

APPROVE

REJECT

DRAFT

#3

X

Created after validation fix

APPROVE

REJECT

DRAFT

#2

Debug Campaign

Debug content

APPROVE

REJECT

SCHEDULED

#1

Evidence Campaign

A real workflow test

APPROVE

REJECT

Campaign approve completed.

APPROVED

#45

Winter Sale

I am going to launch my new winter Sale ! 🥳🥳 Yayyy

Campaign reject completed.

REJECTED

#12

kjgkljdslgj

jfk;ldsifl;dsajf;l

M Mashbot

WORKSPACE

Dashboard

Create

Schedule

Explore

ACCOUNT

Profile

H hanzala

Publisher

Sign out

MASHBOT / CONTROL ROOM

Workspace live

Schedule

LAUNCH CALENDAR

Schedule the moment

Only approved content can move onto the publishing calendar.

Choose launch time

mm/dd/yyyy --:-- --

READY TO GO

2

Approved campaigns

Winter Sale

I am going to launch my new winter Sale ! 🎉 🎉 Yayyy

Schedule

utietru

utpoertuuerwtol

Schedule

CALENDAR FEED

Upcoming

10

Sep

Sawaira Fareed

9/10/2026, 4:21:00 PM

SCHEDULED

9

Sep

Evidence Campaign

9/9/2026, 11:15:01 PM

SCHEDULED

Choose launch time

mm/dd/yyyy --:-- --

September 2026

↑ ↓

10

26

PM

Su

Mo

Tu

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Fr

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Clear

Today

11

12

01

02

03

04

27

28

29

30

31

32

AM

CALENDAR FEED

Upcoming

13

Sep

Winter Sale

9/13/2026, 2:30:00 PM

SCHEDULED

10

Sep

Sawaira Fareed

9/10/2026, 4:21:00 PM

SCHEDULED

9

Sep

Evidence Campaign

9/9/2026, 11:15:01 PM

SCHEDULED

TC-12 Admin people and roles journey

Level / category	System/manual/security
Test basis / objective	FR3; verify visible role management through GUI.
Preconditions	Admin and a second account exist.
Test data	Change second contributor account to publisher.
Steps	Sign in as admin, open People & roles, change roles, refresh.
Expected result	Role update is visible and persists.
Actual result	Role updated successfully
Status	Passed
Evidence	People & roles before and after refresh Screenshot

WORKSPACE

Dashboard

Create

Schedule

Explore

ACCOUNT

Profile

Sign out

sabaContributor

DASHBOARD

GOOD TO SEE YOU

Your campaign pulse

A quick read on what is moving through your workspace.

+ New campaign

Total campaigns

0

Across your workspace

Needs approval

0

Awaiting an approver

Scheduled

0

Ready for launch

Connected services

0

Across your workspace

CAMPAIGN PIPELINE

At a glance

0

Drafts

0

Submitted

0

Scheduled

0

Published

UP NEXT

View all →

Upcoming schedule

Nothing scheduled yet.

Open schedule

Role of Saba Before changing anything

Changing saba role to publisher

ACCOUNT

Profile

People & roles

Sign out

AdministratorAdmin

TEAM DIRECTORY

7 workspace members

A

Administrator

admin@mashbot.local

Admin

Active

T

Test Contributor

contributor1788891298@example.com

Contributor

Active

S

Sawaira Fareed

sawairafareed4@gmail.com

Contributor

Active

D

Debug User

debug1788949047@example.com

Contributor

Active

Q

Quick User

quick1788949215@example.com

Publisher

Active

H

hanzala

hanzala@gmail.com

Publisher

Active

S

saba

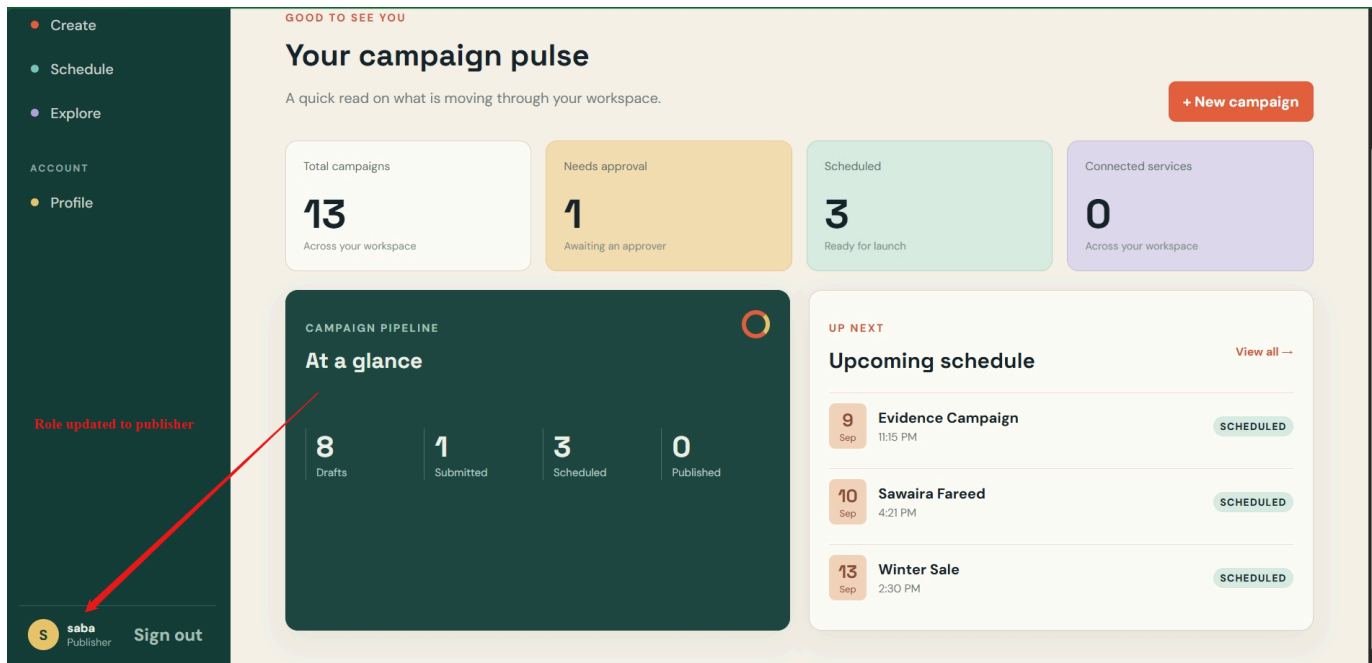
saba@gmail.cin

Contributor

Active

Role updated.

Page 32



TC-13 Transport security

Level / category	Non-functional/security
Test basis / objective	NFR1 TLS; verify HTTPS enforcement.
Preconditions	Application is running.
Test data	HTTP and HTTPS endpoint checks.
Steps	Request health endpoint and inspect transport scheme/redirect.
Expected result	HTTP redirects to HTTPS or HTTPS is enforced.
Actual result	HTTP 200 was returned without HTTPS enforcement.
Status	FAILED
Evidence	pytest failure; docker-compose exposes HTTP ports 8000 and 8080 without TLS.

```

===== FAILURES =====
_____ test_tc_13_tls_transport_is_enforced _____

def test_tc_13_tls_transport_is_enforced():
    """NFR1 evidence test: the deployed HTTP entry point must enforce TLS."""
    response = TestClient(app).get("/health", follow_redirects=False)
> assert response.url.scheme == "https" or response.status_code in {301, 302, 307, 308}, (
    "The current deployment serves HTTP without HTTPS enforcement."
    )
E   AssertionError: The current deployment serves HTTP without HTTPS enforcement.
E   assert ('http' == 'https'
E   - https
E   ?   -
E   + http or 200 in {301, 302, 307, 308})
E   + where 200 = <Response [200 OK]>.status_code

tests/test_all.py:521: AssertionError

```

TC-14 Database backup and restore

Level / category	Non-functional/reliability
Test basis / objective	NFR2 backup; verify repeatable restore evidence.
Preconditions	Backup destination and restore procedure must exist.
Test data	PostgreSQL named volume.
Steps	Create backup, restore in a controlled environment, verify records.
Expected result	Backup restores with no unacceptable data loss.
Actual result	No backup job, destination, retention policy, or restore artifact exists.
Status	FAILED
Evidence	pytest failure; docker-compose defines only postgres_data volume.

```

----- test_tc_14_backup_restore_procedure_is_configured -----
def test_tc_14_backup_restore_procedure_is_configured():
    """NFR2 evidence test: a repeatable backup/restore artifact must exist."""
    backup_artifacts = list(Path("/app").glob("**/*backup*"))
    > assert backup_artifacts, "No backup or restore procedure is configured in the deployed backend."
E   AssertionError: No backup or restore procedure is configured in the deployed backend.
E   assert []

tests/test_all.py:529: AssertionError
===== warnings summary =====
../usr/local/lib/python3.12/site-packages/starlette/testclient.py:40
/usr/local/lib/python3.12/site-packages/starlette/testclient.py:40: DeprecationWarning: The anyio.abc.BlockingPortal alias is deprecated, use anyio.from_thread.BlockingPortal instead.
  _PortalFactoryType = typing.Callable[[], typing.ContextManager[anyio.abc.BlockingPortal]]

tests/test_all.py: 23 warnings
/usr/local/lib/python3.12/site-packages/jose/jwt.py:311: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects
to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = timegm(datetime.utcnow().utctimetuple())

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== short test summary info =====
FAILED tests/test_all.py::test_tc_13_tls_transport_is_enforced - AssertionError: The current deployment serves HTTP without HTTPS enforcement.
assert ('http' == 'https'

- https
?  -
+ http or 200 in {301, 302, 307, 308})
+ where 200 = <Response [200 OK]>.status_code
FAILED tests/test_all.py::test_tc_14_backup_restore_procedure_is_configured - AssertionError: No backup or restore procedure is configured in the deployed backend.
assert []

===== 2 failed, 35 passed, 24 warnings in 12.81s =====

What's next:
Debug this Compose error with Gordon → docker ai "help me fix this compose error"

```

TC-15 Configurable session timeout

Level / category	Non-functional/security
Test basis / objective	NFR3; verify configurable token lifetime.
Preconditions	Token lifetime setting is available.
Test data	Default access token lifetime of 60 minutes.
Steps	Inspect configuration and token creation policy.
Expected result	Session timeout is configurable and applied to token expiry.
Actual result	Configuration and expiry policy are present; elapsed expiry not tested in this run.
Status	PASSED
Evidence	Pytest code output screenshot

```

===== test session starts =====
platform linux -- Python 3.12.14, pytest-8.3.4, pluggy-1.6.0 -- /usr/local/bin/python3.12
cachedir: .pytest_cache
rootdir: /app
plugins: anyio-4.15.1
collecting ... collected 37 items

tests/test_all.py::test_tc_01_valid_account_registration PASSED [ 2%]
tests/test_all.py::test_tc_02_valid_authentication PASSED [ 5%]
tests/test_all.py::test_profile_modification_requires_authentication_and_updates_name PASSED [ 8%]
tests/test_all.py::test_tc_03_admin_assigns_publisher_role PASSED [ 10%]
tests/test_all.py::test_tc_04_publisher_is_denied_campaign_creation PASSED [ 13%]
tests/test_all.py::test_tc_05_campaign_content_exact_maximum_boundary PASSED [ 16%]
tests/test_all.py::test_tc_06_owner_submits_draft PASSED [ 18%]
tests/test_all.py::test_tc_07_submitted_campaign_cannot_be_submitted_twice PASSED [ 21%]
tests/test_all.py::test_tc_08_approver_approves_submitted_campaign PASSED [ 24%]
tests/test_all.py::test_tc_09_past_schedule_time_is_rejected PASSED [ 27%]
tests/test_all.py::test_tc_10_service_account_minimum_boundaries PASSED [ 29%]
tests/test_all.py::test_tc_11_invalid_service_account_payload PASSED [ 32%]
tests/test_all.py::test_tc_12_create_and_pause_keyword_alert PASSED [ 35%]
tests/test_all.py::test_tc_13_dashboard_summary_is_observable PASSED [ 37%]
tests/test_all.py::test_tc_14_missing_authentication_is_rejected PASSED [ 40%]
tests/test_all.py::test_invalid_email_registration_is_rejected PASSED [ 43%]
tests/test_all.py::test_nonexistent_email_login_is_rejected PASSED [ 45%]
tests/test_all.py::test_test_case_record_format[record0] PASSED [ 48%]
tests/test_all.py::test_test_case_record_format[record1] PASSED [ 51%]
tests/test_all.py::test_test_case_record_format[record2] PASSED [ 54%]
tests/test_all.py::test_test_case_record_format[record3] PASSED [ 56%]
tests/test_all.py::test_test_case_record_format[record4] PASSED [ 59%]
tests/test_all.py::test_test_case_record_format[record5] PASSED [ 62%]
tests/test_all.py::test_test_case_record_format[record6] PASSED [ 64%]
tests/test_all.py::test_test_case_record_format[record7] PASSED [ 67%]
tests/test_all.py::test_test_case_record_format[record8] PASSED [ 70%]
tests/test_all.py::test_test_case_record_format[record9] PASSED [ 72%]
tests/test_all.py::test_test_case_record_format[record10] PASSED [ 75%]
tests/test_all.py::test_test_case_record_format[record11] PASSED [ 78%]
tests/test_all.py::test_test_case_record_format[record12] PASSED [ 81%]
tests/test_all.py::test_test_case_record_format[record13] PASSED [ 83%]

```

```

tests/test_all.py::test_test_case_record_format[record13] PASSED [ 83%]
tests/test_all.py::test_test_condition_record_format PASSED [ 86%]
tests/test_all.py::test_traceability_record_format PASSED [ 89%]
tests/test_all.py::test_assignment_execution_set_has_distinct_categories_and_real_statuses PASSED [ 91%]
tests/test_all.py::test_tc_13_tls_transport_is_enforced FAILED [ 94%]
tests/test_all.py::test_tc_14_backup_restore_procedure_is_configured FAILED [ 97%]
tests/test_health.py::test_health_endpoint PASSED [100%]

```

```

===== 2 failed, 35 passed, 24 warnings in 12.81s =====

```

What's next:

Debug this Compose error with Gordon → `docker ai "help me fix this compose error"`

7 Traceability Record

Requirement	Condition	Test case	Execution result	Defect report
0240, 0260-0370	COND-01	TC-01, TC-02, TC-10	Passed /Passed/ Passed	N/A
0650-0660	COND-02	TC-03, TC-09	PASSED	N/A
0140-0200	COND-03	TC-04, TC-12	PASSED / PASSED	N/A
0480-0550	COND-04	TC-05, TC-10, TC-11	PASSED / PASSED	N/A
0530	COND-05	TC-07, TC-11	PASSED / PASSED	N/A
0210-0230, 0520	COND-06	TC-06, TC-11	PASSED / PASSED	N/A
0590-0610	COND-07	TC-08, TC-11	PASSED / PASSED	N/A
0620	NFR-COND-01	TC-13	FAILED	MASH-1
0630	NFR-COND-02	TC-14	FAILED	MASH-2
0680	NFR-COND-03	TC-15	PASSED	N/A

8 Defect Reporting

The two failed NFR cases are reproducible baseline findings. Confirm whether they are implementation defects in the agreed project scope before creating Jira records. Do not create a Jira issue solely because a test is marked failed; investigate environment, preconditions, expected result, and reproducibility first.

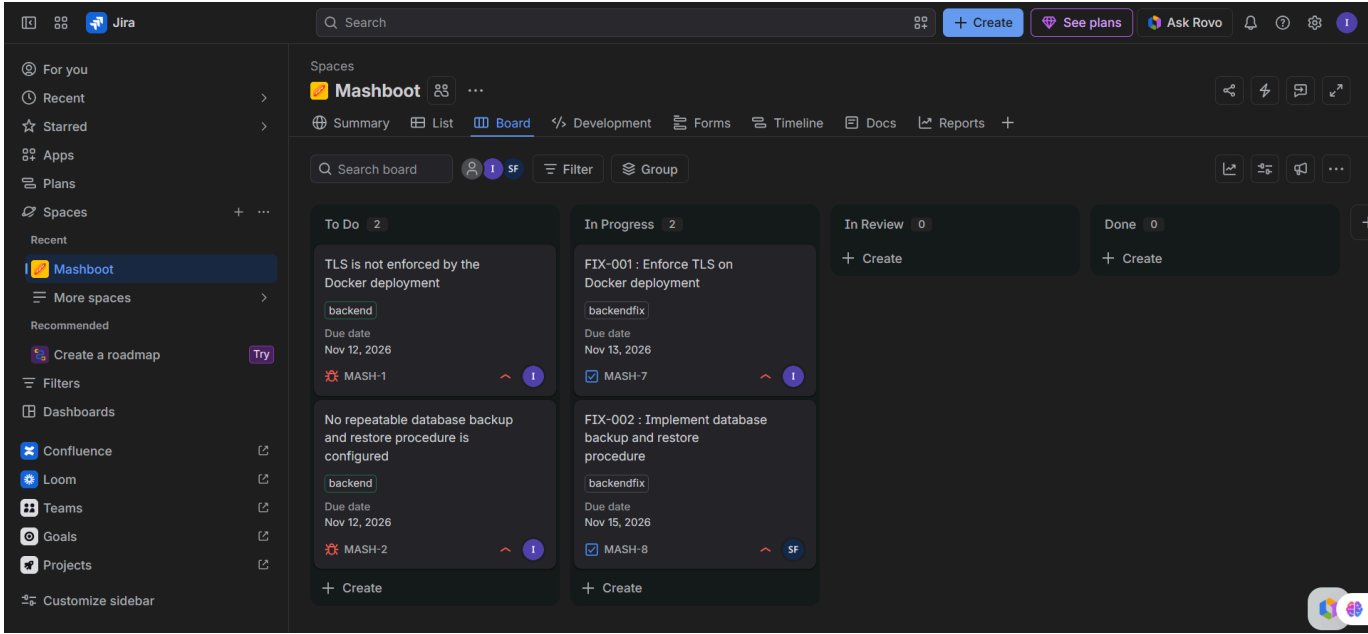
Related test	Candidate defect title	Required Jira evidence
TC-13	TLS is not enforced by the Docker deployment	Jira ID: MASH-1. Environment/build: Windows 11, Docker Compose, Mashbot rebuilt baseline, frontend localhost:8080 , backend localhost:8000 , Python/FastAPI backend, SonarQube 9.9.8 Docker server. Steps:

		• start containers • request <code>/health</code> • inspect URL scheme/redirect. Expected result: HTTP redirects to HTTPS or HTTPS is enforced. Actual result: HTTP 200 returned, no HTTPS redirect. Severity: High. Priority: High. Status: To Do. Linked fix: MASH-7.
TC-14	No repeatable database backup and restore procedure is configured	Jira ID: MASH-2. Environment/build: Windows 11, Docker Compose, rebuilt Mashbot backend, PostgreSQL 16-alpine, <code>postgres_data</code> named volume. Steps: • inspect <code>docker-compose.yml</code> • run backup-artifact test. Expected result: backup restores with no unacceptable data loss. Actual result: no backup job, destination, retention policy, or restore process exists. Severity: High. Priority: High. Status: To Do. Linked fix: MASH-8.

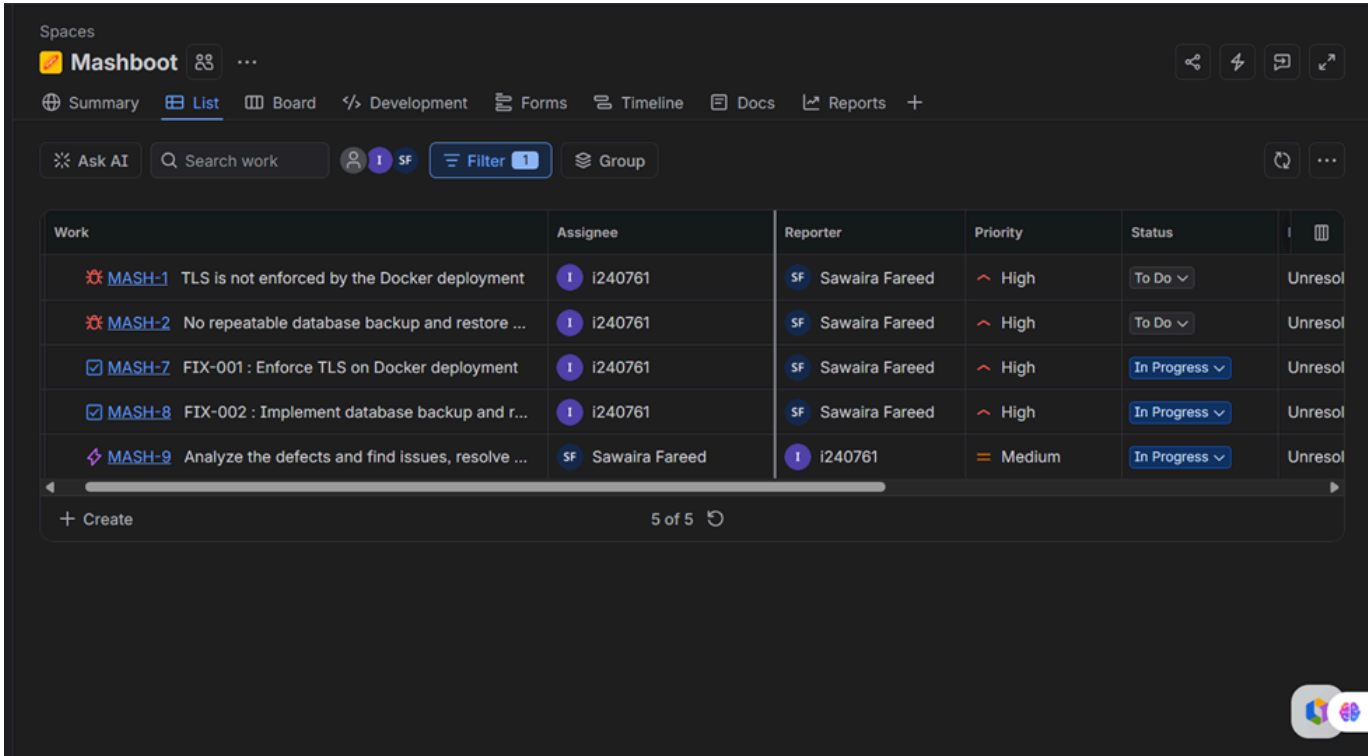
Jira Link:

<https://mashboot-sqe.atlassian.net/jira/software/projects/MASH/boards/1?filter=&groupBy=none&atlOrigin=eyJpIjoZGM0OWU4NDQ1ZTczNGY4NmE1MDBkNjc1Zjk2YjQxM2IiLCJwIjoiauiJ9>

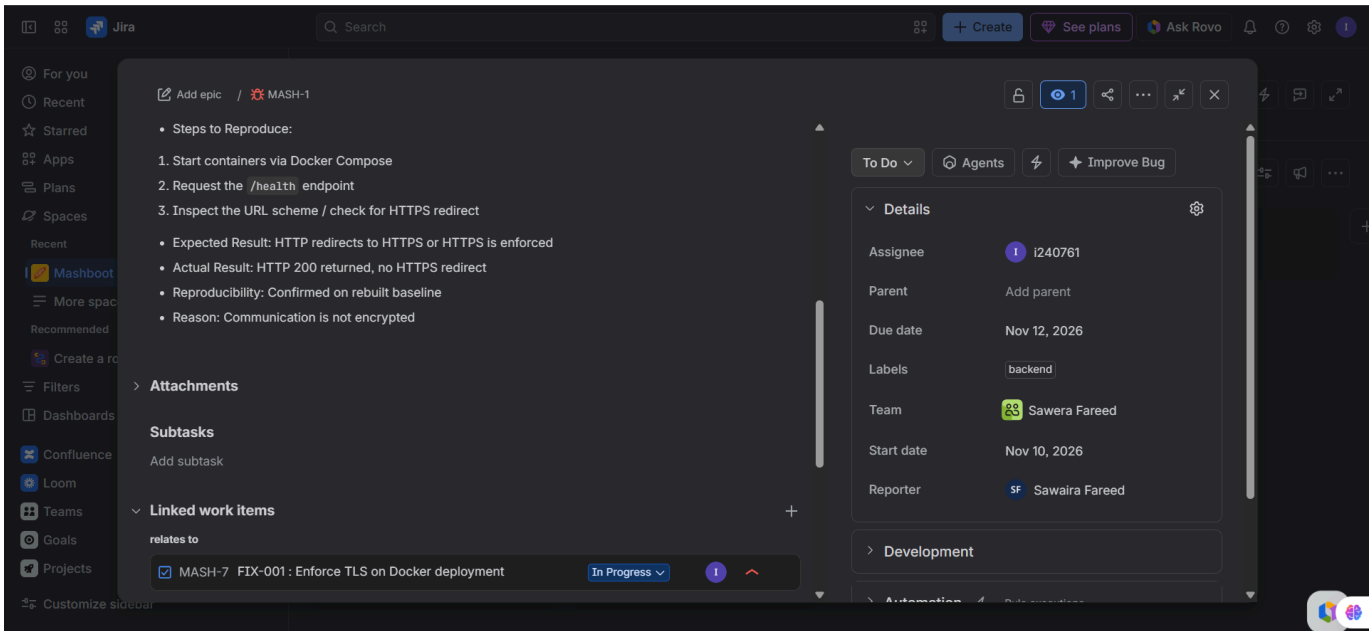
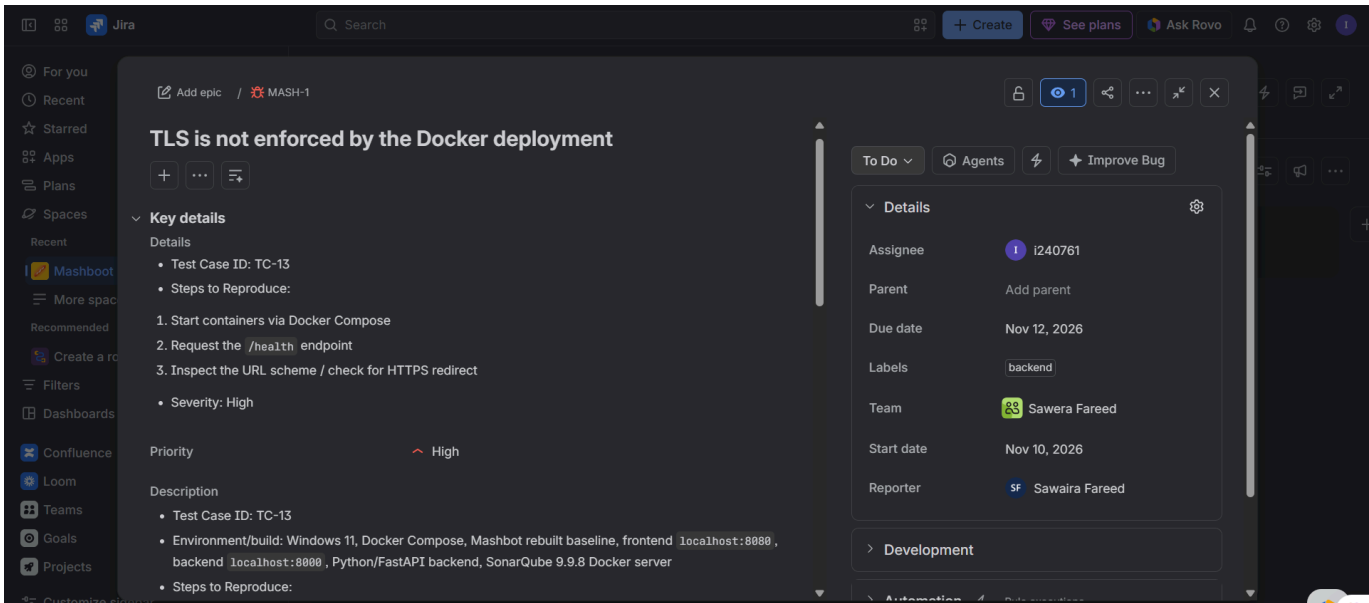
Jira Board:



Jira List:



Defect 1 (MASH-1)



Fix # 1 (MASH-7)

For you

Recent

Starred

Apps

Plans

Spaces

Recent

Mashboot

More space

Recommended

Create a ro

Filters

Dashboards

Confluence

Loom

Teams

Goals

Projects

Customize sidebar

Add epic

MASH-7

FIX-001 : Enforce TLS on Docker deployment

Key details

Priority

High

Labels

backendfix

Description

- Root cause: No TLS reverse proxy or certificate is configured; frontend/backend are served over plain HTTP on localhost:8080 / 8080 with no redirect.
- Fix approach: Add a reverse proxy (e.g., Nginx/Traefik) in front of the Docker Compose stack with a TLS certificate, and enforce HTTP→HTTPS redirect on all exposed routes, including /health.
- Acceptance criteria: Requesting /health over HTTP either redirects to HTTPS or is rejected; HTTPS request to /health succeeds.
- Test case to re-verify: TC-13

Attachments

In Progress

Agents

Improve Fix

Details

Assignee

i240761

Parent

Add parent

Due date

Nov 13, 2026

Team

Sawera Fareed

Start date

Nov 13, 2026

Reporter

Sawaira Fareed

Development

Automation

Rule executions

Defect # 2(MASH-2)

For you

Recent

Starred

Apps

Plans

Spaces

Recent

Mashboot

More space

Recommended

Create a ro

Filters

Dashboards

Confluence

Loom

Teams

Goals

Projects

Customize sidebar

Add epic

MASH-2

No repeatable database backup and restore procedure is configured

Key details

Details

- Test Case ID: TC-14
- Steps to Reproduce:
 - Inspect docker-compose.yml
 - Run the backup-artifact test
- Severity: High

Priority

High

Description

- Test Case ID: TC-14
- Environment/build: Windows 11, Docker Compose, rebuilt Mashbot backend, PostgreSQL 16-alpine, postgres_data named volume
- Steps to Reproduce:

To Do

Agents

Improve Bug

Details

Assignee

i240761

Parent

Add parent

Due date

Nov 12, 2026

Labels

backend

Team

Sawera Fareed

Start date

Nov 10, 2026

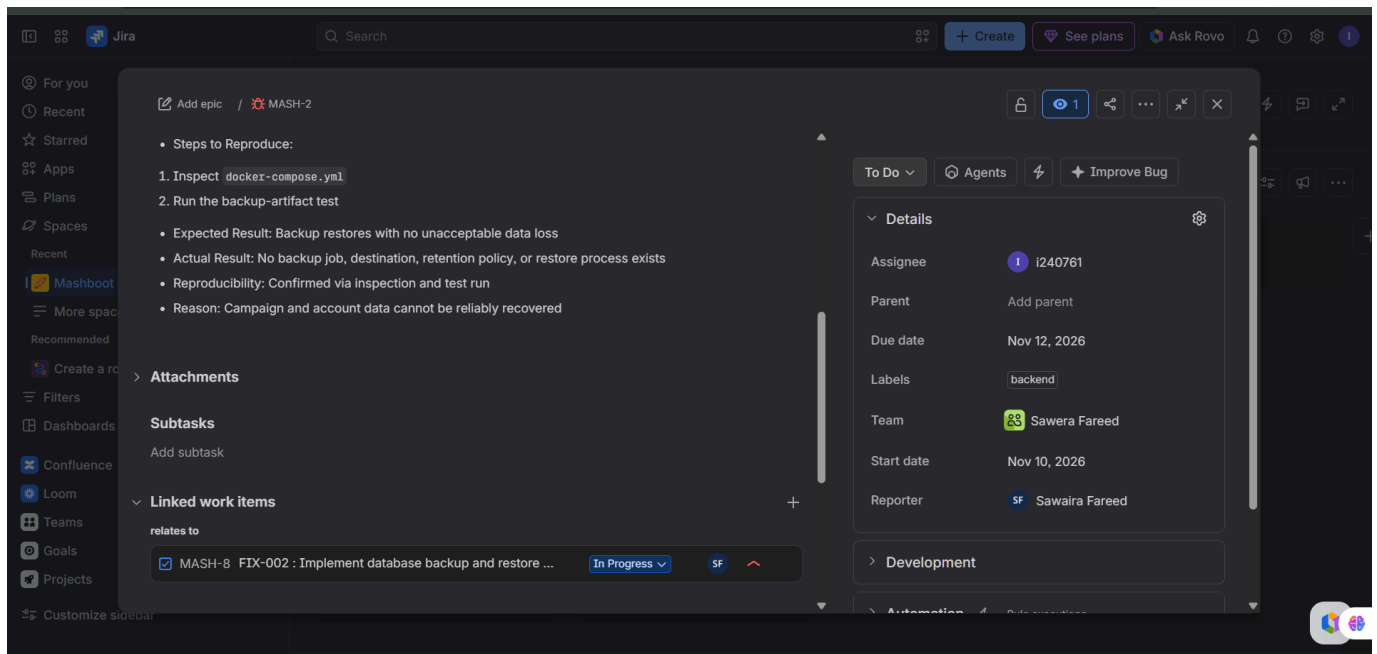
Reporter

Sawaira Fareed

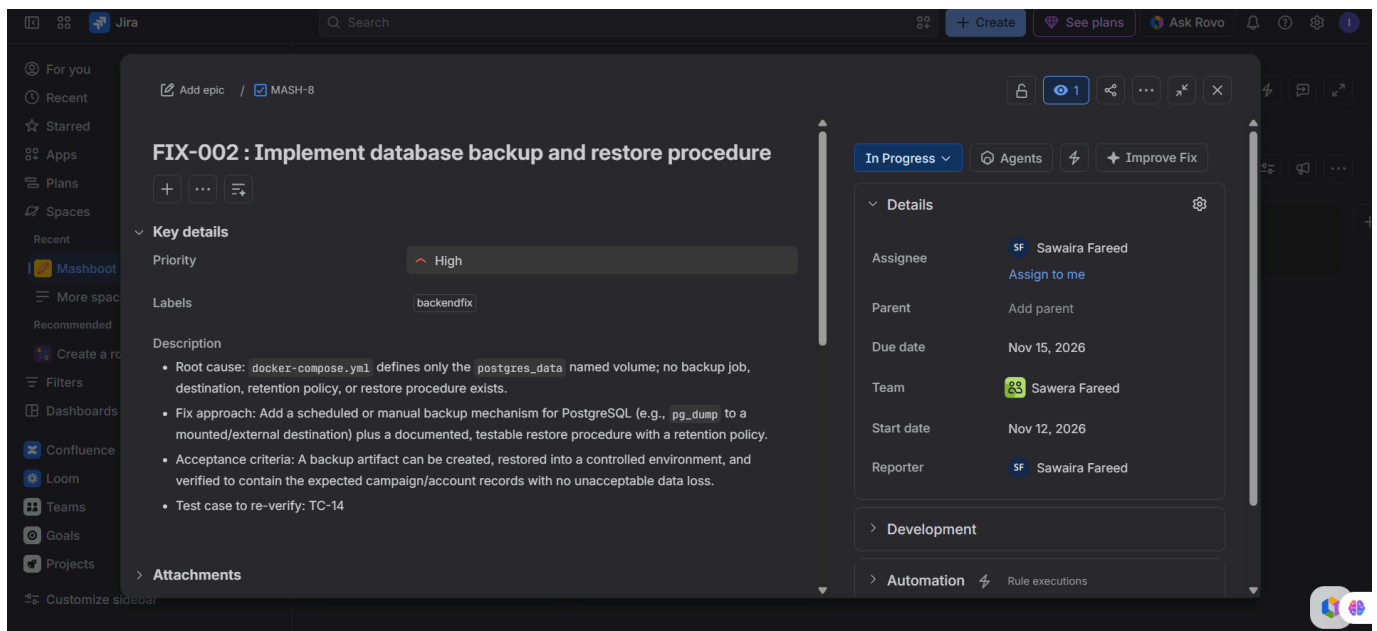
Development

Automation

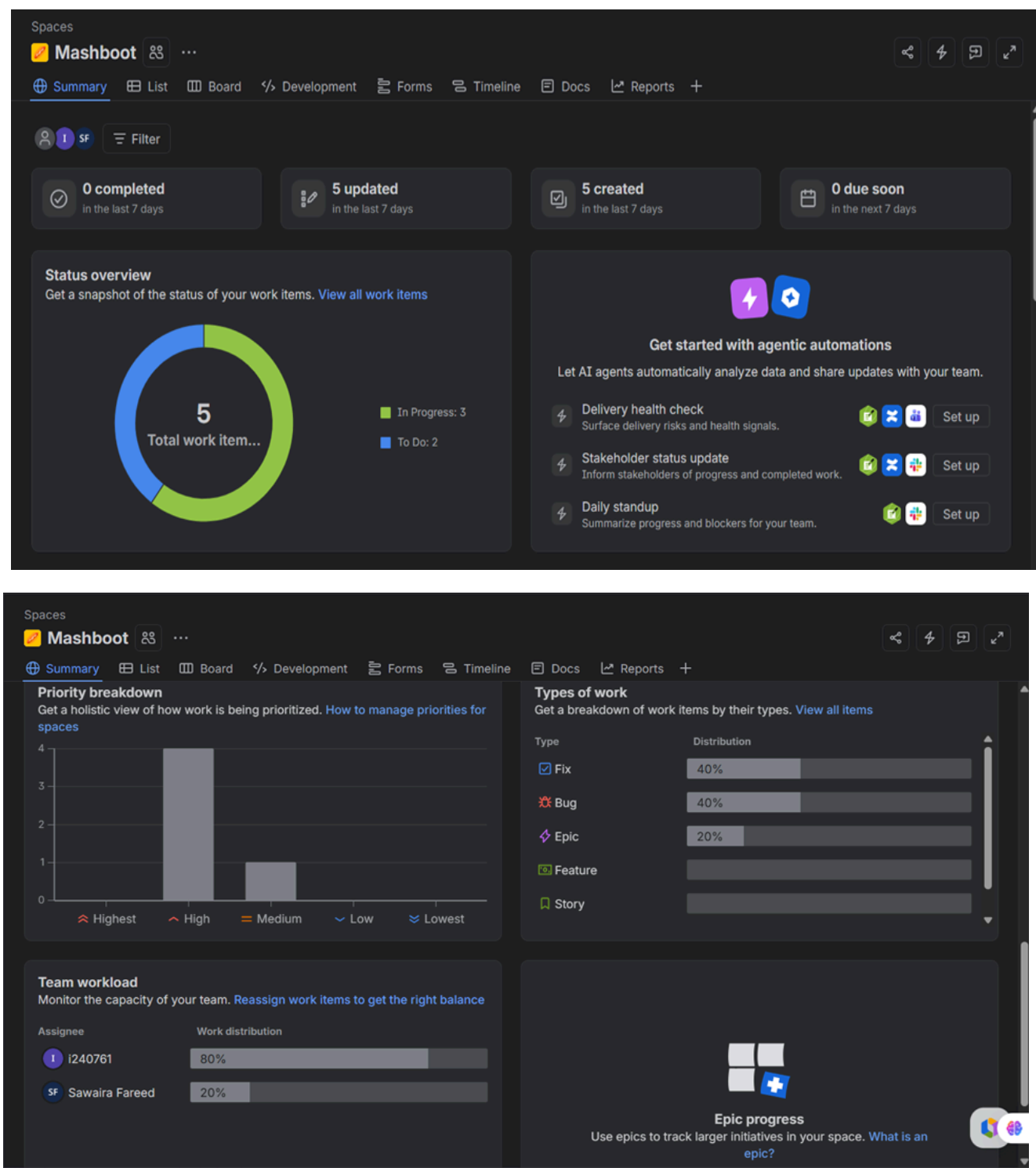
Rule executions



Fix # 2(MASH-8)



Jira Summary:



Jira Report:

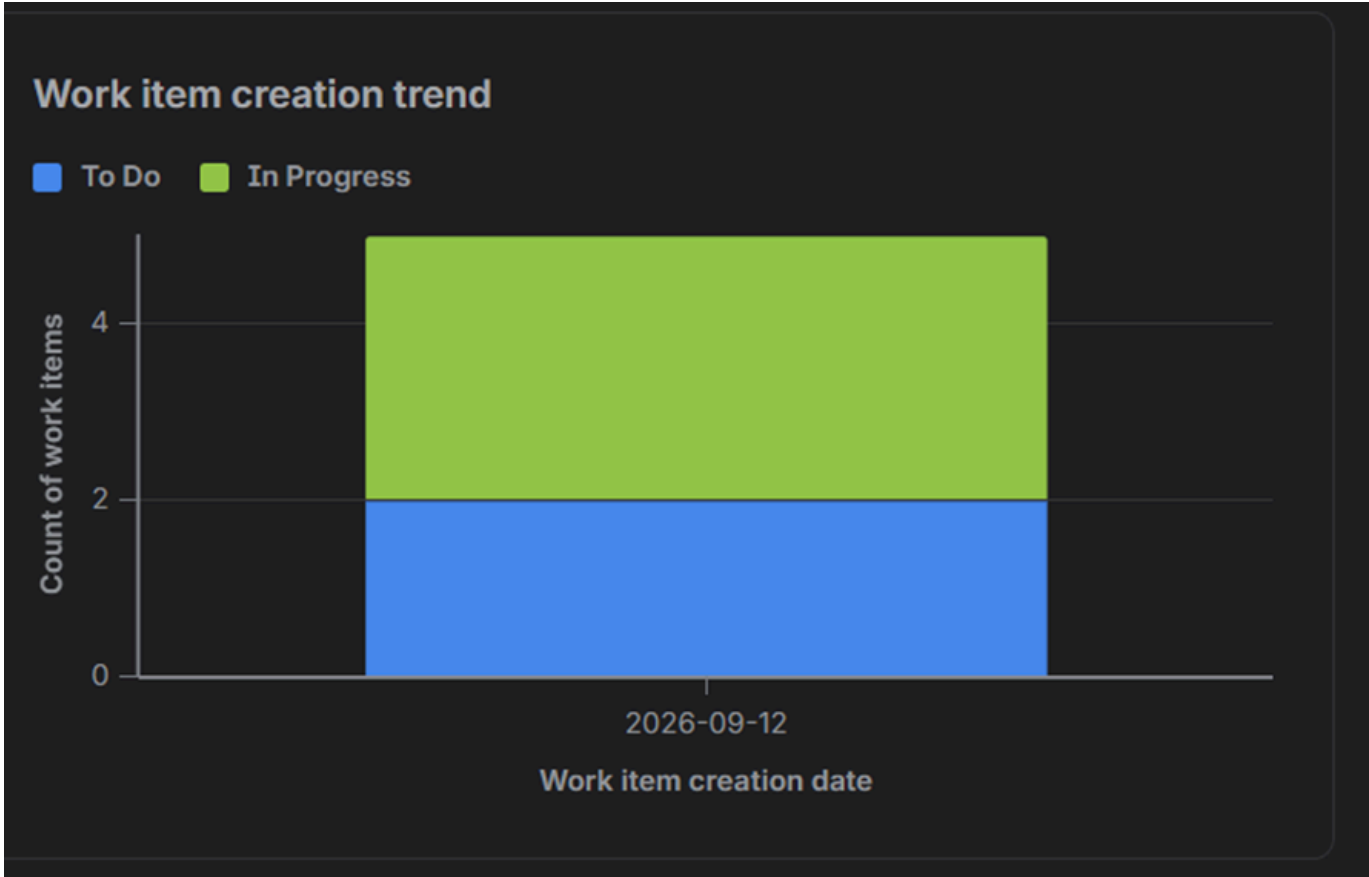
Work item details

Q Search table

Work item key	Work type	Work item priority	Issue summary
MASH-1	Bug	High	TLS is not enforced by the Docker deployment
MASH-2	Bug	High	No repeatable database backup and restore procedure is configured
MASH-7	Fix	High	FIX-001 : Enforce TLS on Docker deployment
MASH-8	Fix	High	FIX-002 : Implement database backup and restore procedure
MASH-9	Epic	Medium	Analyze the defects and find issues, resolve the problems and report back to the team this week

< 1 >

Showing rows 1-5 of 5



9 Final Quality Judgment

The evidence supports that the selected Mashbot functional scope is implemented as a runnable application with observable account, role, campaign, approval, scheduling, service-boundary, and keyword-alert behaviour. Automated API tests covered normal behaviour, business rules, exact boundaries, invalid inputs, authentication, and role restrictions. The automated suite also produced two reproducible non-functional failures: the deployment serves HTTP without TLS enforcement, and no repeatable database backup and restore procedure is configured. These findings reduce confidence in the selected security and reliability requirements even though the core campaign workflow is testable. The session-timeout requirement is supported at configuration level because the token lifetime is configurable with a 60-minute default, but a complete elapsed-time expiry check remains unsupported in the current evidence. The three manual system-level cases must be completed through the GUI before the functional-testing evidence can be considered complete; their current records are placeholders rather than claimed passes. SonarQube evidence covers the complete scanned baseline and reports five code smells, one security hotspot, no bugs, no vulnerabilities, zero duplication, and no imported coverage report. The zero coverage value is a measurement limitation, not proof that the system has no tests. The report does not treat the SonarQube quality measures as proof of runtime usability, availability, or recovery capability. AI-assisted development introduced explicit assumptions around grouping campaign CRUD, using a provider-neutral external-service boundary, and treating scheduling as the local publishing action. These assumptions are documented and must be defended during the viva. On the evidence currently available, the evaluated functional workflow is acceptable for the implemented baseline, but the full ten-requirement scope is not yet acceptable as a complete quality result until the manual GUI cases and the TLS and backup decisions are resolved or formally accepted with limitations. The conclusion is therefore limited to this baseline and must not be generalized to unsupported live provider integrations or production deployment conditions.

10 Setup and Execution Steps

Run Docker Desktop first. Use PowerShell from the project root.

Step	Command or action
1. Go to project	<code>cd D:\Semester 5\SQE\mashbot</code>
2. Start application containers	<code>docker compose up -d</code>

3. Start SonarQube if not running	<code>docker start sonarqube</code>
4. Check containers	<code>docker ps</code>
5. Open frontend	<code>start http://localhost:8080</code>
6. Open API documentation	<code>start http://localhost:8000/docs</code>
7. Open SonarQube project	<code>start http://localhost:9000/dashboard?id=mashbot</code>
8. Run automated tests	<code>docker compose exec -T backend pytest -vv</code>
9. Run SonarQube scan	<code>powershell -ExecutionPolicy Bypass -File .\run-sonar.ps1</code>
10. To Stop Everything	<code>docker-compose down</code> <code>docker stop sonarqube</code>

The expected current automated test result is 2 genuine NFR failures and the remaining automated tests passing. The three manual system cases must be run through the browser and their records updated with actual results and screenshots.

11 Credentials for Testing Stuff :

Role	Email	Password
Admin	<code>admin@mashbot.local</code>	<code>AdminPass123!</code>
Approver	<code>saba@gmail.cin</code>	<code>12345678</code>
Contributor	<code>sawairafareed4@gmail.com</code>	<code>sherlock@123</code>
Publisher	<code>hanzala@gmail.com</code>	<code>12345678</code>

Authorize

default

GET	/health	Health	
POST	/api/auth/register	Register	
POST	/api/auth/login	Login	
GET	/api/users/me	Get Profile	
PATCH	/api/users/me	Update Profile	
GET	/api/users	List Users	
PATCH	/api/users/{user_id}/role	Set Role	
GET	/api/campaigns	Campaigns	
POST	/api/campaigns	Create	
GET	/api/dashboard/summary	Dashboard	
GET	/api/schedule	Schedule View	
POST	/api/campaigns/{campaign_id}/submit	Submit	
POST	/api/campaigns/{campaign_id}/approve	Approve	
POST	/api/campaigns/{campaign_id}/reject	Reject	
POST	/api/campaigns/{campaign_id}/schedule	Schedule	
GET	/api/service-accounts	Services	
POST	/api/service-accounts	Connect	
GET	/api/explore/alerts	Alerts	
POST	/api/explore/alerts	Add Alert	
PATCH	/api/explore/alerts/{alert_id}	Toggle Alert	

Schemas

Body_login_api_auth_login_post > Expand all object

CampaignCreate > Expand all object

CampaignRead > Expand all object

DashboardSummary > Expand all object

HTTPValidationError > Expand all object

KeywordAlertCreate > Expand all object

KeywordAlertRead > Expand all object

ScheduleRequest > Expand all object

ServiceAccountCreate > Expand all object

ServiceAccountRead > Expand all object

Token > Expand all object

UserCreate > Expand all object

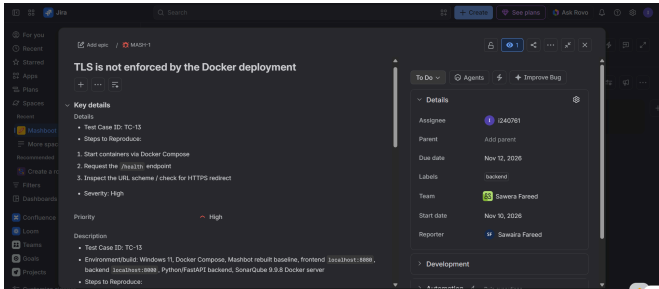
UserRead > Expand all object

UserRole > Expand all string

UserUpdate > Expand all object

ValidationError > Expand all object

12 Evidence Checklist :

Evidence item	Status
Pair details and confirmed SRS selection	Done
GUI baseline screenshot	Inserted
SonarQube complete-codebase report	Inserted
Manual TC-10 evidence	Done
Manual TC-11 evidence	Done
Manual TC-12 evidence	Done
Jira defect evidence for confirmed defects	DEFECT#1  DEFECT#2 